Platform plan: Cosmos, Blob, Service Bus, and the apps / module / resources layout

Status: **agreed approach**, updated 2026-11-08 (evening). Cosmos for **NoSQL** is a client requirement. §7 shows what is built per phase; §12 lists what is still missing. Settings for Cosmos, Service Bus and the API key are in `core-pipeline/.env` (values never in chat).

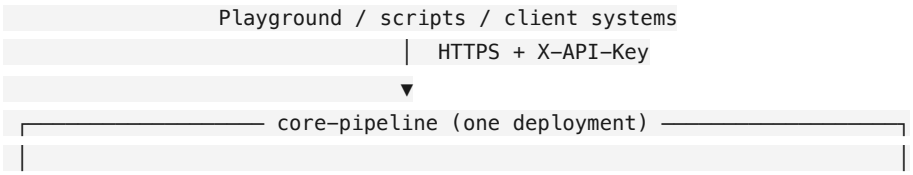
Built so far (Postgres stays the default; Cosmos and Service Bus are built and off until switched on):

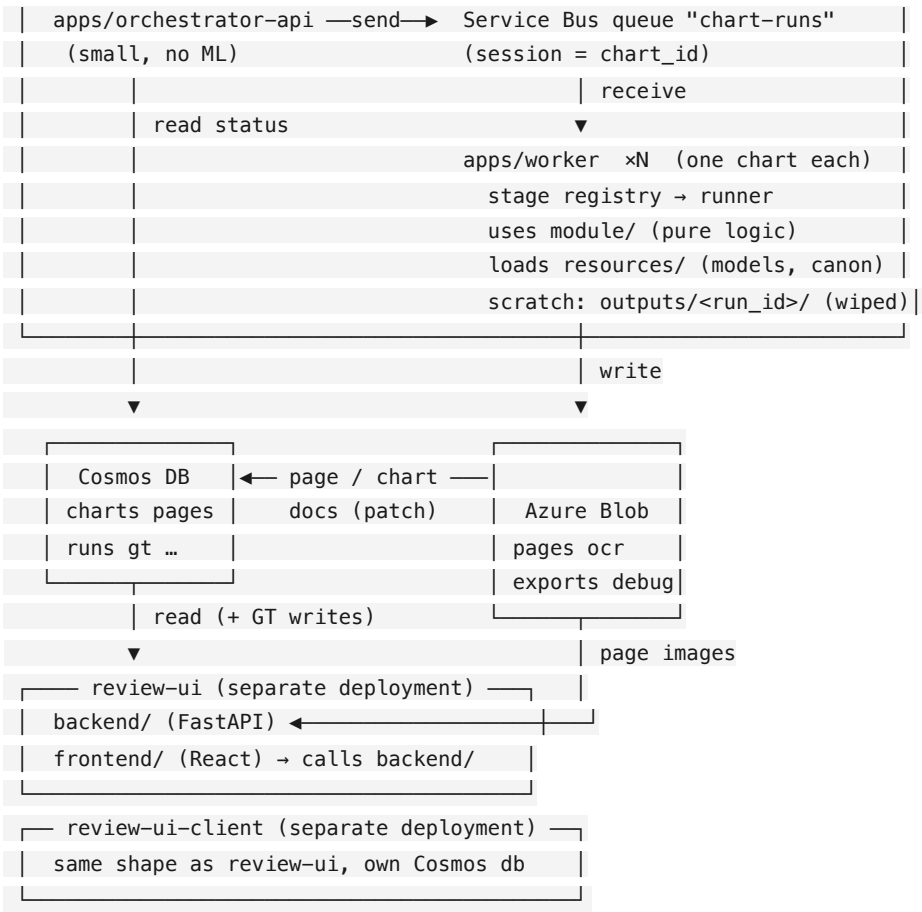
- **Layout (phase 1):** core-pipeline `apps/orchestrator_api/`, `apps/worker/`, `common/`, `modules/`, `resources/canon/`; review-ui `backend/app/{routers,store}/` and a frontend with its own `BACKEND_URL`. About 3,600 lines of dead code removed, and more on 2026-11-07 (old blank/junk training code, the unused sequencing cross-encoder).
- **review-ui reads only the database and Blob.** No Local mode, no File Viewer.
- **`core-pipeline/outputs/` is scratch:** a chart folder not run for 7 days is deleted. Logs are deleted after 7 days.
- **One schema file** (`schema/schema.sql`) plus `schema/migrate.sql` for existing databases.
- **`DB_BACKEND=postgres|cosmos|both`** is in both `.env` files and wired: `cosmos` runs on Cosmos, `both` writes Postgres and copies every write to Cosmos (reads stay on Postgres). `RUN_MODE=queue` sends chart runs through Service Bus. See §7 and §12.
- **db-utilities** (new, 2026-11-07): manifests and ground truth are loaded by their own Docker job, not by core-pipeline (§3).
- **Monitoring on Postgres:** a `run_usage` row per chart run, daily and hourly views, `python cli.py usage` (§5a).
- **`deploy/`:** the GitHub → Azure DevOps copy leaves out `training/`, `scripts/`, `tests/`.

1. Goals

1. Move the database from Postgres to **Azure Cosmos DB** (NoSQL API, provisioned throughput).
2. Keep page images, OCR JSON and exports in **Azure Blob**. Blob is kept forever for now.
3. Manage run requests through **Azure Service Bus** queues.
4. Reorganise core-pipeline into `apps/orchestrator_api`, `apps/worker`, `modules/` and `resources/`. review-ui keeps `frontend/` and `backend/`, each deployable on its own. All Python and TypeScript (no .NET).
5. **core-pipeline, review-ui and review-ui-client are separate infrastructure.** review-ui's frontend and backend can also be deployed apart. Each has its own code, `.env`, Dockerfiles and `infra/`. They share no code at runtime, only the data in Cosmos and Blob.
6. **Models and keyword canons are resources, not code.** They live in `core-pipeline/resources/` and are versioned and replaced without a code release.
7. Write just enough: one source of truth per fact. Debug files are written only in `--debug` mode, **to both** local disk (`core-pipeline/outputs/debug/`) and Blob (`debug/`). A debug run still writes all its results to Cosmos as usual.
8. **No more output to `review-ui/data`.** `core-pipeline/outputs/` is temporary scratch. **review-ui reads only Cosmos and Blob; Local mode is removed.**
9. Step-by-step APIs (OCR, page type, member verification, DOS) that run only the stages they still need.
10. **Monitoring:** every run records time taken, Doc Intel pages and cost, and LLM tokens and cost (once an LLM is used), in one place with daily and hourly totals.
11. **Reference data is loaded by its own job.** Manifests and ground truth go into the database through **db-utilities**, a separate Docker job, never through core-pipeline or review-ui.
12. **Models come from Blob,** not from a manual copy (phase 8), so every host and every pod gets identical weights.
13. **Hosting stays on the Ubuntu VM.** AKS is parked (2026-11-07); every part already runs as a container, so it stays possible later.

2. Overall approach





A run, step by step:

1. A caller posts to the orchestrator: a full run, or a target such as `/member-verify`. The orchestrator checks the API key, writes a `runs` doc, puts one message on `chart-runs` and returns `202 + run_id`.
2. A worker takes the message. Service Bus sessions guarantee that one chart is processed by only one worker at a time.
3. The worker works out which stages the target needs and skips those already done with the same `version` and `input_hash`. It runs the rest.
4. Each stage reads its inputs from Cosmos and Blob, runs `module/` code with `resources/` loaded, and writes its result as one patch on the page doc. Large JSON goes to Blob.
5. The worker marks the run done and completes the message. Its scratch folder `outputs/<run_id>/` is deleted. With `debug`, the explanation files are also written to `outputs/debug/<run_id>/` on that machine's disk and to Blob `debug/<run_id>/`.
6. review-ui shows the chart from Cosmos, with images from Blob. Ground truth saves go to Cosmos.

3. Target structure: separate deployments

```
advantmed-imaging-pipeline/              (one GitHub repo; three deployables)
├── core-pipeline/                      ── DEPLOYMENT 1 ──
│   ├── apps/
│   │   ├── orchestrator_api/           FastAPI: auth, enqueue, status, exports
│   │   │   ├── app/{main.py,routers/,services/}
│   │   │   ├── requirements.txt
│   │   │   └── Dockerfile
│   │   └── worker/                     Service Bus consumer, runs the chain
│   │       ├── worker/{main.py,pipeline/,stages/}
│   │       ├── requirements.txt        (torch, docling, gliner … only here)
│   │       └── Dockerfile
│   ├── modules/                        the logic (was stages/lib; moved 2026-11-07)
│   │   ├── image_preprocess/ ocr/ blank_junk/ page_classify/ extraction/
```

```
│   │   └── member/ dos/ continuity/ encounter/ sequencing/ imaging_final/
│   ├── common/                          config, logging, db/ (today); cosmos/, blob/, servicebus/,
│   │                                    artifacts.py, contracts/ (doc models)
│   ├── resources/                       local cache of Blob resources/ (git-ignored)
│   │   ├── manifest.json                the live versions, as last read from Blob
│   │   ├── models/                      weights, loaded at worker start
│   │   │   ├── hw/ rapidocr/ docling/ semantic-model/ ner/
│   │   │   └── blank-junk/ page-family/ encounter/ kv-extraction/
│   │   └── canon/                       keyword JSONs, checked before each run
│   ├── outputs/                         scratch per run (wiped) + debug/ (debug mode only)
│   ├── logs/                            7-day retention
│   ├── infra/                           Cosmos db+containers, SB queue (Bicep)
│   ├── tests/
│   ├── docker-compose.yml               orchestrator-api + worker ×N
│   └── .env / .env.example
│
├── review-ui/                           ── DEPLOYMENTS 2a + 2b ──
│   ├── backend/                         2a: FastAPI, deployable on its own
│   │   ├── app/
│   │   │   ├── main.py
│   │   │   ├── routers/                 charts, pages, ground_truth, accuracy, users …
│   │   │   ├── services/                accuracy, excel_compare, labels (no queries)
│   │   │   ├── store/                   postgres/ and cosmos/ (DB_BACKEND), blob.py
│   │   │   └── contracts/               copy of core-pipeline's doc schemas
│   │   ├── tests/  requirements.txt  Dockerfile  .env / .env.example
│   ├── frontend/                        2b: React, static build, deployable on its own
│   │   ├── src/{pages,components,api,lib}/
│   │   ├── package.json  Dockerfile (nginx)  .env.example (API URL)
│   ├── infra/                           one file per part (backend, frontend)
│   └── docker-compose.yml               both together, for one-host installs
│
├── review-ui-client/                    ── DEPLOYMENT 3 ── (ui-temp repo, synced)
│
├── db-utilities/                        ── DEPLOYMENT 4 ── (a job, not a server; built)
│   ├── dbutil/                          manifest, metadata-extra, ground-truth, sequence
│   ├── Dockerfile  docker-compose.yml   `docker compose run --rm db-utilities <job> …`
│   ├── input/                           files to load (read-only at /input), or a Blob prefix
│   └── .env / .env.example              DATABASE_URL (later Cosmos) + AZURE_STORAGE_*
│
├── schema/                            Postgres: schema.sql, migrate.sql, migrations/
├── deploy/                            GitHub → DevOps copy, client .gitignore
├── training/  scripts/  tests/        GitHub only, never on the VM
├── utilities/   docs/
└── README.md
```

Rules

- **`modules/` never touches I/O.** It takes text, boxes and settings in, and returns results.
- **Only `worker/stages/` adapters read and write storage,** always through `imaging_common`.
- **`orchestrator_api` never imports `modules`** and has no ML dependencies, so it stays small and starts fast.
- **Code never hard-codes a model path or a canon file.** It asks `resources` for `models/page-leaf` or `canon/continuity_canon` and gets the version from `manifest.json`. The version is stamped on every stage result.
- **The deployments share data, not code.** core-pipeline owns the Cosmos document shapes (`imaging_common/contracts`, pydantic). Their JSON Schemas are copied into `review-ui/backend/app/contracts/`, and a test fails when the two differ. `utilities/sync_client_ui.sh` keeps doing the copy for review-ui-client.
- **Nothing writes into another deployment's folder.** The `review-ui/data/folders` hand-off has ended (built).

- **review-ui's frontend talks only to its backend**, over HTTP at the URL set at build time. It never reads the database or Blob itself, so the two parts can run on different hosts, or the frontend from static hosting.

Resources: models and canon (from Blob)

Not built yet (phase 8). On the VM today the weights are copied by hand into `core-pipeline/models/`.

Both live in Blob as versioned folders in the container **`imaging-models`** (2026-11-07). A small `manifest.json` says which version is live.

```
imaging-models/                          (Blob container)
├── manifest.json                        item -> live version + file fingerprints
├── models/<name>/<version>/…            e.g. models/page-leaf/v3/
└── canon/<version>/*.json               e.g. canon/2026-11-07a/
```

	Models	**Canon**
What	trained weights (BERT encoders, GLiNER, RapidOCR, Docling, MiniLM, handwriting)	keyword JSONs: section headers, continuity, duplicates, leaf keywords …
Source of truth	Blob only (too big for git)	git, so tests run on it and every edit has a history; `utilities/publish_resources.py` uploads a version to Blob
When a worker reads it	**once, when the worker starts**	**before each chart run** (one small manifest read; downloads only on a change)
Local copy	`core-pipeline/resources/` on the host, shared by every worker on it	same
Changing one	upload a new version folder and change `manifest.json`; takes effect on the next restart, never mid-batch	publish or upload, change `manifest.json`; live for the next chart, no redeploy
Rolling back	point `manifest.json` at the old version	same

- **A half-finished upload can't be used:** the worker checks each file against the fingerprint in the manifest and refuses a mismatch.
- **Every result records its versions.** Each stage stamps the model and canon versions it ran with into `input_hash`, so a canon change re-runs exactly the stages it affects.
- **If Blob can't be reached,** the worker keeps what's already on disk and records that in `/health` and in the run record.
- **What it removes:** copying model weights to the VM by hand (the first worker start downloads about 3 GB, once), and the zip → DevOps → VM round trip for a canon edit.
- **Weights are never built into a Docker image:** an image would be many GB and every model change would need a new image.

Where today's code goes

Today	Moves to
`core-pipeline/modules/*` (logic)	`core-pipeline/modules/*`
`core-pipeline/resources/canon/`	git copy stays in the repo; published to Blob `resources/canon/`
`core-pipeline/models/`	Blob `imaging-models/models/`, cached in `core-pipeline/resources/models/`
`core-pipeline/stages/*` runners, `core-pipeline/apps/worker/runner.py`	`core-pipeline/apps/worker/worker/{stages,pipeline}/`
`core-pipeline/apps/orchestrator_api/main.py`	`core-pipeline/apps/orchestrator-api/app/` (split into routers)
`core-pipeline/db/*`, `blob_store.py`, `paths.py`	`core-pipeline/common/imaging_common/{cosmos,blob}/`
`review-ui/data/folders` (pipeline output)	**done:** scratch in `core-pipeline/outputs/charts`; results from the database + Blob
`review-ui/backend/`	stays; split into `routers/`, `services/`, `store/`
`review-ui/frontend/`	stays; own Dockerfile and API URL, deployable alone
review-ui Local mode (`DATA_MODE=local`, disk CSV/JSON readers)	**done:** removed; the disk readers still inside `store/local` go with the `store/` split
`review-ui/.../adapters/postgres/repository.py`	`review-ui/backend/app/store/postgres/`, beside `store/cosmos/`
`schema/*.sql`	`imaging_common/contracts` + `core-pipeline/infra/`

4. Data: Postgres → Cosmos, disk → Blob, jobs → Service Bus

4.1 Cosmos (NoSQL API, provisioned throughput)

Created 2026-11-07: account `medicalimagingtest` (South India, free tier), database `imaging` at **1,000 RU/s manual, shared by its containers**. No containers yet: they are created with the Cosmos code (phase 2), because a partition key cannot be changed later.

Partition key: `/chart_name` for the chart-level containers (proposed 2026-11-07, was `/chart_id`). The chart name is the client's record id: it is known before a chart is ingested, and manifests and ground truth are already keyed on it. A Postgres `chart_id` has no Cosmos equivalent. `users` uses `/id`; `metrics` uses `/period` (`day:2026-11-07`).

Container	Document	Replaces
`charts`	One doc per chart: manifest info, status, per-stage state, summary, member verdict, tags	`chart_list`, `chart_status`, member summary, run stats
`pages`	One doc per page, with one sub-object per stage (`quality`, `ocr` refs and text, `blank_junk`, `classification`, `kv`, `member`, `dos`, `continuity`, `sequence`, `final`)	~20 per-page result tables, plus the staging JSONs
`runs`	One doc per run: request, stages run, timings, errors	`pipeline_jobs`, `page_stage_status`, run logs
`ground_truth`	One doc per page, plus a chart-level doc	the ground-truth tables
`users` (pk `/id`), `manifest` (pk `/chart_name`, the record id), `metrics` (pk `/period`)	—	users, manifest, accuracy snapshots

- **How core-pipeline writes (built 2026-11-07, `common/db/cosmos_store.py`):** the process keeps each chart it is working on in memory (the tested `MemoryStore` logic), loads it from Cosmos on first use, and after every write upserts only the documents that changed (one page doc, plus the chart doc). Ids come from a counter document taken in blocks. One chart is worked on by one process at a time, which keeps this safe; a patch per stage sub-object is a later optimisation.

- **Stage completion** is recorded as `stages.<name> = {done_at, version, input_hash}`. A redelivered message is then a no-op. `skip_ocr` and the step-by-step APIs read the same record.
- **Size limit:** a document can be at most 2 MB. Word boxes and the full Docling / Azure JSON go to Blob; the page doc keeps the text and the blob path.
- **Throughput (free tier):** 1,000 RU/s manual at the database level, shared by all its containers, and 25 GB of storage free. Batch runs may hit the limit (HTTP 429); the SDK retries, and the worker count keeps writes under it. Raise it (or switch to autoscale) from the metrics when it throttles; anything over 1,000 RU/s or 25 GB is billed.
- **Indexing:** large text (page OCR text, kv candidates) is excluded from the index, so each write costs fewer RUs.
- **Network: private endpoint** (2026-11-07). Cosmos is reachable from the VM and from the Mac, so the Cosmos code is developed and checked from the Mac as well. The unit tests use an in-memory store behind the same interface.
- **review-ui-client gets a replica:** its own database in the same account (for example `imaging_client`, created when the client copy moves to Cosmos: a second database's throughput is outside the free 1,000 RU/s), with the same containers, filled from the main database by the clone job (today's `utilities/clone_client_db.py`, rewritten for Cosmos): the chosen charts, users, and no ground truth. It never reads the main database.
- **History is migrated** (§7, phase 7). Every Postgres chart, page result, run, ground truth and user is copied into Cosmos by a one-off utility, and counts are compared before cutover.

4.1a Who writes what

Workers write to the database directly. Nothing routes their results through the orchestrator.

Writer	Cosmos	Blob
Worker	page docs (one patch per stage), chart status and summary, its `runs` doc (stage in progress, timings, errors)	pages, OCR JSON, exports, debug files
Orchestrator API	creates the `runs` doc; reads status	— (reads exports for downloads)
review-ui	ground truth, reviews, users, chart tags	— (reads page images)
db-utilities (jobs)	`manifest` (members per record), `ground_truth` (bulk loads)	— (reads files to load)
Migration utility (once)	everything copied from Postgres	—

Each writer has its own key: workers and the orchestrator get read-write access, review-ui gets read-write only where it saves, and reports read with a read-only key.

4.2 Blob (everything kept for now)

Container	Holds
`inbox`	the existing raw input (unchanged)
`pages`	`<chart>/<page>.png`, rotated / corrected pages: what review-ui shows
`ocr`	`<chart>/<page>.{prelim,final1,final2}.json`: what `skip_ocr` reuses and the overlays come from
`exports`	final CSV / JSON deliverables
`debug`	`<run_id>/…`, written only with `debug: true` (a copy of what goes to local disk)
`imaging-models`	model weights and canon by name and version, plus `manifest.json` (§3)

There are no lifecycle rules for now. If storage cost becomes an issue, one rule on `debug` is the only change needed.

4.3 Service Bus

- **Queue `chart-runs`, sessions on, with the session id set to `chart_id`.** Two runs of one chart are processed in order and never at the same time. Different charts run in parallel. Sessions need the **Standard** tier or higher.
- **The message** is `{run_id, chart_id, target, only?, skip_ocr, debug, input}`.
- **Long charts:** the worker renews the message lock while it works, because one lock lasts at most 5 minutes. If a worker dies, the message comes back and finished stages are skipped. After 5 failed deliveries the message moves to the dead-letter queue, and the `runs` doc says so.

- **Retired:** the Postgres job table, the lease sweeper and FastAPI background tasks.

4.4 The DB toggle (short term)

Until Postgres is retired, `.env` picks the store (built: the setting is checked at start and shown in `/health`; only `postgres` is wired so far):

```
DB_BACKEND=postgres     # today's schema/schema.sql database (default)
DB_BACKEND=both         # write Postgres AND Cosmos, read Postgres: the side-by-side week
DB_BACKEND=cosmos       # the containers in §4.1
```

- core-pipeline, review-ui and db-utilities each read their own `DB_BACKEND`. They must agree, or review-ui shows charts the pipeline is no longer writing.
- Stages, routes and jobs call one interface (`store.charts`, `store.pages`, `store.runs`, `store.ground_truth`, `store.manifest`, `store.users`). Only the implementations know SQL or Cosmos.
- `both` makes every write go to both stores, so the week before cutover compares them on real runs. Reads stay on Postgres until the switch to `cosmos`.
- Switching moves no data. The history migration (phase 7) copies it, and a mismatch check runs before the default changes to `cosmos`.

4.4a What Cosmos for NoSQL changes in the code

Cosmos for NoSQL has no joins across containers and no SQL views, so every query that joins tables today is rewritten around the documents in §4.1:

Today (Postgres)	On Cosmos
core-pipeline `common/db/` (about 2,000 lines of SQL helpers)	`store/cosmos/`: read and patch the chart and page documents
review-ui `store/postgres/repository.py` and the services with SQL (accuracy, value search, Manual Review, review log)	`store/cosmos/`: chart / page reads by partition; cross-chart pages read **precomputed** docs in `metrics`
Report views (`consolidated_chart_results`, `v_run_usage_*`)	documents written when a run ends, rebuilt by a nightly job
`page_stage_status` rows	`stages.<name>` on the page doc (§4.1)
db-utilities upserts	upsert into `manifest` / `ground_truth`

This is the largest single piece of work in the plan (phases 2, 6 and 7).

4.5 Auth

The orchestrator API takes an **API key** in the `X-API-Key` header, from `ORCHESTRATOR_API_KEY`. `/health` stays open. review-ui keeps its own sign-in, and its Playground sends the key when it starts runs.

5. Write just enough, and `--debug`

	Written	Where
Always	Page and chart docs	Cosmos
Always	Page images, OCR JSON, final export	Blob (`pages`, `ocr`, `exports`)
During a run only	Working files	`core-pipeline/outputs/<run_id>/`, deleted when the run ends
Only with `debug: true`	`staging/*.json` explanations, per-stage CSVs, the scratch folder	**Both**: local disk `core-pipeline/outputs/debug/<run_id>/` on the worker (deleted after 7 days, like the logs) and Blob `debug/<run_id>/` (kept)
Logs	`core-pipeline/logs/`	deleted after 7 days (built)

- **CSVs on demand:** `GET /v1/charts/{id}/export?format=csv&stage=…` builds the CSV from Cosmos when asked.
- **One writer:** `artifacts.write(kind, name, data)` is the only file writer. A test fails if a stage writes a file directly.
- **Data that stages pass to each other moves into the page doc.** Today `staging/extraction.json` and `staging/rotation.json` carry it; those become Cosmos fields.

5a. Monitoring: time, Doc Intel cost, LLM tokens

Today: `core-pipeline/logs/run_stats.jsonl` gets one line per chart run, with time taken, pages and the number of pages sent to Doc Intel. `run_stats_charts.csv` and `run_stats_hourly.csv` are rebuilt from it, and `python cli.py stats` prints the hourly table. It lives in the log folder, so it's lost when a VM is rebuilt, and it records no cost or tokens.

Target: every billable or timed thing is recorded where it happens and kept in the database, never only in a log file.

What	Recorded by	Fields
Time	the worker, per stage and per run	started, ended, seconds; pages; seconds per page; queue wait (enqueued → started)
Doc Intel	the Final OCR 2 stage, per page call	pages sent, pages billed, model (`prebuilt-layout` …), features, retries, throttles (429s), cost
LLM (when one is used)	one wrapper around every LLM call	provider, deployment, prompt tokens, completion tokens, cached tokens, latency, cost, which stage and page
Errors	the worker	failed stage, retry count, dead-lettered runs

Where it is written:

- **Per run:** a `usage` block on the `runs` document (Cosmos), written as the run goes, so a crash still leaves what was spent.
- **Totals:** a `metrics` container holding one document per day and per hour, with charts, pages, seconds, Doc Intel pages and cost, and LLM tokens and cost, split by trigger, stage and model. It's updated by the worker when a run ends and rebuilt by a nightly job, so a missed update fixes itself.
- **While Postgres is still in use (built):** the same rows go to the `run_usage` table (Part 1 of `schema.sql`), with `v_run_usage_daily` and `v_run_usage_hourly`. Prices: `.env` section 5.
- **Costs come from a price table, not code:** `PRICE_DOCINTEL_PER_1000_PAGES` and `PRICE_LLM_<deployment>_IN` / `_OUT` per 1,000 tokens in `.env`. The price used is stored with each row, so a later price change never rewrites history.

Reading it:

- `GET /v1/metrics?from=&to=&group=day|hour|trigger|stage` on the orchestrator API.
- `python cli.py stats` (the same numbers, in the terminal).
- A **Usage** page in review-ui for admins: time per chart, Doc Intel pages and cost per day, and tokens per stage.
- **Alerts** (optional): a daily Doc Intel spend above a set amount, or a run slower than a set time, sends a message through the `run-events` topic.

6. Step-by-step APIs

One stage registry with dependencies, for example `member_verify requires [ocr_final2, kv_extract, blank_junk_2]`. Each endpoint names a target. The worker runs only what is missing or stale for that target, then stops.

Endpoint	Runs up to	Returns
`POST /v1/charts` (full)	stage 15	`run_id`
`POST /v1/charts/{id}/ocr`	rotation → prelim → final1 → final2 → section headers	per-page text, engine, quality
`POST /v1/charts/{id}/page-type`	OCR + blank/junk 1 and 2 + kv + page classification + subtype	per-page type / subtype / blank-junk
`POST /v1/charts/{id}/member-verify`	OCR + blank/junk + kv + member	member verdict, wrong-member pages
`POST /v1/charts/{id}/dos`	… + DOS	per-page and per-document dates
`GET /v1/runs/{run_id}`	—	status, the stage in progress, errors, the result when done

How a caller gets the answer. This was decision 3, and it is now set to the default below. A 400-page chart takes minutes, longer than an HTTP request should stay open. So every endpoint replies at once with `202` and a `run_id`, and the caller asks `GET /v1/runs/{run_id}` until it says done; that reply carries the result. For a small chart, or a target already done, add `?wait=60`: the call waits up to 60 seconds and returns the result directly if it finishes in time. Tell me if a caller needs something else.

- `only`, `skip_ocr` and `debug` work as today.
- **Re-running:** a stage counts as done only if its `version` and `input_hash` match. The hash includes the model and canon versions, so a canon edit re-runs exactly the stages it affects.

7. Phases

#	Phase	What	Done when	Status (2026-11-08)
0	Access	Cosmos, Service Bus and Blob details in `core-pipeline/.env`; `infra/` (Bicep) creates the database, containers and queue	the code connects to all three from the Mac and from the VM	**Mostly:** settings in `.env`; queue `chart-runs`; database `imaging` (1,000 RU/s) and its containers created; Blob `imaging-models`. `infra/` (Bicep) written, not yet run
1	Restructure, no logic change	`apps/`, `modules/`, `common/`, `resources/`; review-ui `backend/` + `frontend/` each with its own Dockerfile	same outputs on the test charts; tests pass	**Done** except: the API is still one `main.py` (not split into routers), and the API and worker are one image, one process
2	Storage layer + DB toggle	one data-access interface, Postgres and Cosmos implementations; `DB_BACKEND=postgres|cosmos|both`	both pass the same tests; one chart end to end on each	**Built for core-pipeline and db-utilities** (2026-11-07): `common/db` + `CosmosStore`, `both` mirror, a full test chart on Cosmos (outputs match Postgres except number formatting and one page type: a classifier tie, fixed 2026-11-08 with `tie_priority`); db-utilities `--to postgres|cosmos|both`. review-ui: chart views only
3	Outputs and debug	every file written inventoried; stage-to-stage data into page docs; `artifacts.write`; scratch wiped per run; debug to disk and Blob	a non-debug run writes only to the DB and Blob	**Partly:** scratch with a 7-day purge; debug goes to local disk only; `staging/*.json` still carries data between stages
4	Service Bus + API key	orchestrator enqueues, workers consume (sessions, lock renewal, dead-letter); `X-API-Key`	kill a worker mid-chart and the chart finishes on another	**Built 2026-11-08, off by default** (`RUN_MODE=queue`, `API_KEY_REQUIRED=true`): `apps/worker/queue_worker.py`, compose `worker` (`--scale worker=3`), `GET /api/runs/{id}?wait=`; Playground sends the key. Checked with a smoke message on the real queue. Not yet: the kill-a-worker test on the VM, separate API / worker images
5	Step-by-step APIs	stage graph, version / input-hash check, `/v1/...` endpoints	`/member-verify` runs only what it needs; a second call is instant	**Built 2026-11-08:** `/v1/charts/run`, `/v1/charts/{name}/ocr`, `/page-type`, `/member-verify`, `/dos`, `/v1/runs/{id}?wait=`; only the stages not yet completed run, a repeat call answers at once. Not yet: version / input-hash staleness (use `rerun: true`)
6	review-ui on Cosmos	`store/cosmos/`, precomputed accuracy docs; sync review-ui-client	the test charts look the same as today	**Built 2026-11-08** (`DB_BACKEND=cosmos` / `both`): chart views, landing list, ground truth (read + save), users and sign-in, review log, tags, encounters, delete, OCR search, Manual Review. Accuracy report and search by value on Cosmos since 2026-11-08 (chart by chart / narrow cross-partition reads, matched as on Postgres). Nothing in review-ui is 501 on Cosmos any more. Synced to review-ui-client 2026-11-08
7	History migration	`utilities/migrate_postgres_to_cosmos.py`; `DB_BACKEND=both` for a week; db-utilities writes	counts match;	**Migration built** (2026-11-08): `utilities/migrate_postgres_to_cosmos.py`,

#	Phase	What	Done when	Status (2026-11-08)
	and cutover	Cosmos; Cosmos becomes the default	Postgres read-only, then retired	the Mac's 5 charts and its user copied, counts match. Not yet: the run on the VM's database, the week on `both`, the cutover
7a	Monitoring	`run_usage` now; `runs.usage` + `metrics` on Cosmos; LLM wrapper; `/v1/metrics`; review-ui Usage page	every run shows time, Doc Intel pages and cost	**Partly:** `run_usage` and `cli.py usage` built; no LLM wrapper, `/v1/metrics` or Usage page
8	Resources from Blob	`manifest.json`, `publish_resources.py`, models fetched at worker start, canon checked before each run, fingerprints	a new model reaches every host with no manual copy	**Built 2026-11-08, off by default** (`RESOURCES_FROM_BLOB`): `common/resources.py` fetch with sha256 check, `utilities/publish_resources.py`, `/health.resources`; checked on the real `imaging-models` container. Not yet: canon re-check before each chart, versions stamped on results
9	AKS	—	—	**Parked** (2026-11-07)
—	db-utilities	manifests and ground truth loaded by their own Docker job	core-pipeline has no load code	**Done** for Postgres, Cosmos or both (`--to`, `load.sh` / `load.ps1`)

Phase 1 goes first because every later change then lands in its final place, and its outputs can be compared one for one.

8. Running and deploying

Development happens on the Mac. On the client side the code arrives as a **zip downloaded from GitHub on the Windows laptop** (no git there), goes to **Azure DevOps**, and is deployed on the **Ubuntu VM (16 CPU, 64 GB RAM)**. Every round trip is slow, so each phase ships as one zip with every change for that phase.

Windows laptop (PowerShell): refresh the code. `deploy/prepare_for_devops.ps1` copies the GitHub zip into the DevOps folder without `training/`, `scripts/` and `tests/`, keeps `.env` files and weights, and installs `deploy/devops.gitignore`. Full steps are in `deploy/README.md`.

```
powershell -ExecutionPolicy Bypass -File "$env:USERPROFILE\Downloads\model-repo-main\deploy\prepare_for_devops.ps1" `
    -Source "$env:USERPROFILE\Downloads\model-repo-main" -Target "C:\work\advantmed-devops"
# then push C:\work\advantmed-devops to Azure DevOps as usual
```

Ubuntu VM (bash): deploy core-pipeline (each deployment is started from its own folder)

```
cd ~/advantmed-imaging-pipeline/core-pipeline
# after pulling the new code from Azure DevOps
docker compose build
docker compose up -d --scale worker=3     # 3 workers × 1 chart each
curl -fsS localhost:8001/health
docker compose logs -f worker
```

Ubuntu VM (bash): deploy review-ui (both parts on one host; each can also run alone)

```
cd ~/advantmed-imaging-pipeline/review-ui
docker compose up -d --build                  # backend + frontend
docker compose up -d --build backend          # backend only
docker compose up -d --build frontend         # frontend only (API URL set in its .env)
```

Sizing the 16 CPU / 64 GB VM. One worker holds Docling, GLiNER and the BERT models at about 10–14 GB, and uses 4 to 5 CPUs well. Three workers plus the orchestrator API and review-ui fit with headroom. Start at `--scale worker=3` and go to 4 only if memory stays under 48 GB during a batch. Because jobs come from Service Bus, adding a worker on another machine later needs no code change.

9. Decisions (2026-11-07)

#	Question	Decision
1	Cosmos throughput	**Provisioned** at the database level (see 23: 1,000 RU/s on the free tier)
2	History	**Migrate everything** from Postgres
3	Step-by-step APIs: sync or async	**Default chosen:** async with an optional `?wait=` (§6)
4	Blob retention	**Keep everything for now**, debug included. Debug files go to both local disk and Blob
5	Orchestrator API auth	**API key** (`X-API-Key`)
6	Models and canon	**Resources inside core-pipeline** (`resources/`), outside the code
7	Deployments	**core-pipeline, review-ui and review-ui-client are separate infra**
8	Pipeline output	`core-pipeline/outputs/` is temporary; **review-ui is DB only, no Local mode**
9	Short term	**A `DB_BACKEND=postgres|cosmos` toggle** in core-pipeline and review-ui until Postgres is retired
10	Client copy	`training/`, `scripts/`, `tests/` stay on GitHub only; DevOps and the VM get the runtime (`deploy/`)
11	Logic folder name	`core-pipeline/modules/` (not `model/`), imported as `modules.<concern>` (built 2026-11-07)
12	review-ui layout	`frontend/` and `backend/` stay, each deployable on its own; the frontend talks only to the backend
13	Models and canon delivery	Blob `resources/` with `manifest.json`: models loaded at worker start, canon checked before each run; canon also kept in git
14	Debug files	Written to both local disk and Blob `debug/`
15	Monitoring	Time, Doc Intel pages and cost, LLM tokens and cost: recorded per run and kept as daily and hourly totals in the database (§5a)
16	Database	**Cosmos for NoSQL**: a client requirement (evening of 2026-11-07)
17	DB toggle values	`postgres`, `cosmos`, or `both` (write both, read Postgres)
18	Manifests and ground truth	Loaded only by **db-utilities**, a separate Docker job; not part of core-pipeline or review-ui
19	Models	Blob container **`imaging-models`**, fetched at start (phase 8); never built into images
20	Hosting	The Ubuntu VM; **AKS parked**
21	Cosmos network	**Private endpoint**; reachable from the VM and the Mac
23	Cosmos throughput	**Free tier: 1,000 RU/s manual**, shared by the containers of `imaging` (created); 25 GB free
24	review-ui key	The **same read-write key**: review-ui edits ground truth
22	review-ui-client data	A **replica database** (same containers) in the same Cosmos account, filled by the clone job

10. What I need from you

Already in `core-pipeline/.env`: the Cosmos endpoint, key and database; the Service Bus connection string and queue; the API key; the Blob account. Still open:

Cosmos DB

- Done: database `imaging` at 1,000 RU/s; review-ui uses the same read-write key.
- Agree the partition key `/chart_name` (§4.1) before the containers are created.

Blob

- Permission to create the `imaging-models` container, or who creates it?
- On the VM: a managed identity (preferred) or a key?

11. Risks

Risk	Answer
Cosmos cost on review-ui's cross-chart accuracy pages	Precomputed metric docs in `metrics`
The 2 MB document limit	OCR JSON and word boxes stay in Blob
Duplicate deliveries	Stage completion is recorded per stage, so a redelivered message skips finished work
Worker memory with GLiNER, Docling and BERT together	One chart per worker; at most 3–4 workers on the 64 GB VM
Slow client round trip (zip → DevOps → VM)	One zip per phase; models from Blob (phase 8) remove the manual copy
History migration misses data	The migration utility compares counts per table against Cosmos and refuses cutover on a mismatch
review-ui and core-pipeline drift apart on document shapes	Contract copy plus a test that fails on a difference

12. What is still missing (updated 2026-11-08)

Done since 2026-11-07: the storage layer in core-pipeline (`common/db`, 69 functions, contract test); `CosmosStore` (`DB_BACKEND=cosmos`), checked with a full test chart run on the real database; `DB_BACKEND=both` (Postgres plus a copy of every write in Cosmos, same ids); db-utilities writing manifests and ground truth to Postgres, Cosmos or both (`load.sh`); review-ui's chart views (detail, OCR, imaging, page images) read from Cosmos.

In build order:

1. **History migration (phase 7):** built (`utilities/migrate_postgres_to_cosmos.py`, the Mac's 5 charts and its user copied with matching counts); left: the run on the VM's database.
2. **review-ui on Cosmos (phase 6):** built and synced to review-ui-client, the Accuracy View (and its Download Page Level Accuracy) included since 2026-11-08: it reads each chart's documents, so with many charts precomputed `metrics` docs will be needed for speed and RU. Search by value works on Cosmos too (2026-11-08): it reads every page's Final values and ground truth per search, the same scaling caveat.
3. **Models and canon from Blob (phase 8):** built (off by default); left: the canon re-check before each chart and model / canon versions stamped on results.
4. **Service Bus and separate workers (phase 4):** built (off by default); left: the kill-a-worker test on the VM, and a slim API image without the models.
5. **Outputs (phase 3).** Stage-to-stage data out of `staging/*.json` into the page docs; scratch wiped after each run; debug files to Blob as well as disk.
6. **Step-by-step APIs (phase 5):** built; left: version / input-hash staleness.
7. **Monitoring leftovers (7a).** LLM token wrapper, `/v1/metrics`, the review-ui Usage page.
8. **review-ui-client replica.** The clone job rewritten for Cosmos (a replica database).
9. **`infra/`:** written (Bicep for Cosmos, the queue, `imaging-models`); not yet run (no Azure CLI on the Mac).
10. **Postgres-only utilities:** `utilities/pull_client_ground_truth.py` (the client's ground truth into ours, 2026-11-08) and `utilities/tag_chart.py` (mark charts, e.g. "Special") write Postgres only; with `DB_BACKEND=both`, re-run the migration for those charts or add a Cosmos path. How to run them is below.

Running the two utilities

Both use the core-pipeline venv and `DATABASE_URL` from `core-pipeline/.env`.

Mark charts (tags, e.g. "Special"): `utilities/tag_chart.py`. A chart is named by its chart name or chart_list id. review-ui's History and Accuracy pages filter by tag.

```
# Ubuntu VM
cd ~/advantmed-imaging-pipeline/core-pipeline && source .venv/bin/activate
python ../utilities/tag_chart.py 52743839_44976074 --tag Special          # add a tag
python ../utilities/tag_chart.py --from-file charts.txt --tag Special      # one chart per line
python ../utilities/tag_chart.py 52743839_44976074 --remove Special        # remove it
python ../utilities/tag_chart.py 52743839_44976074                         # show its tags
python ../utilities/tag_chart.py --list-tags                               # every tag in use

# Windows (PowerShell)
cd core-pipeline; .venv\Scripts\Activate.ps1
python ..\utilities\tag_chart.py 52743839_44976074 --tag Special
python ..\utilities\tag_chart.py --from-file charts.txt --tag Special
python ..\utilities\tag_chart.py --list-tags
```

The client's ground truth into ours: `utilities/pull_client_ground_truth.py`. Reads the client's review database (`CLIENT_DATABASE_URL`, read only) and writes ours (`DATABASE_URL`). For every chart the client has ground truth on, ours is replaced by theirs (page labels and history, member decision, encounters, Reviewed mark, Review Log); other charts keep ours. `--apply` first backs up all six tables (schema `gt_backup_<stamp>` in our database, plus CSVs in `utilities/output/gt_backup_<stamp>/`). Without `--apply` it is a dry run.

```
# Ubuntu VM
cd ~/advantmed-imaging-pipeline/core-pipeline && source .venv/bin/activate
export CLIENT_DATABASE_URL='postgresql://...the client database...'
python ../utilities/pull_client_ground_truth.py                      # dry run: counts per table
python ../utilities/pull_client_ground_truth.py --apply              # back up, then copy
python ../utilities/pull_client_ground_truth.py --apply --charts-file charts.txt   # only these
python ../utilities/pull_client_ground_truth.py --apply --add-users   # client accounts too
(inactive)
python ../utilities/pull_client_ground_truth.py --list-backups
python ../utilities/pull_client_ground_truth.py --restore gt_backup_20261108_093000   # undo

# Windows (PowerShell)
cd core-pipeline; .venv\Scripts\Activate.ps1
$env:CLIENT_DATABASE_URL = "postgresql://...the client database..."
python ..\utilities\pull_client_ground_truth.py
python ..\utilities\pull_client_ground_truth.py --apply
python ..\utilities\pull_client_ground_truth.py --restore gt_backup_20261108_093000
```